

Towards guidelines and verification methods for automated vehicle HMIs



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ABSTRACT

In most levels of vehicle automation, drivers will not be merely occupants or passengers of automated vehicles. Especially in lower levels of automation, where the driver is still required to serve as a fallback level (SAE L3) or even as a supervisor (SAE L2), there is a need to communicate relevant system states (e.g., that the automated driving system works reliably or that there is a need for manual intervention) via the Human-Machine Interface (HMI). However, there are currently no agreed-upon guidelines that apply specifically to HMIs for automated driving. In this paper, we summarize design recommendations for visual-auditory and visual-vibrotactile HMIs derived from empirical research, applicable standards and design guidelines pertaining to in-vehicle interfaces. On this basis, we derive an initial set of principles and criteria for guiding the development and design of automated vehicle HMIs. A heuristic evaluation methodology consisting of an itemized checklist evaluation that can be used to verify that basic HMI requirements formulated in the guidelines are met is also presented. The heuristic evaluation involves an inspection of the HMI during typical use cases, judging their compliance with the proposed guidelines and documentation of identified instances of non-compliance. Taken together, the combination of the proposed guidelines and the heuristic evaluation methodology form the basis for both design and validation recommendations of automated vehicle HMIs, that can serve the industry in the important evolution of automation within vehicles.

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1. Introduction

Automated driving will profoundly change the role of the driver from actively controlling the vehicle to supervising its actions (so-called “partial automation”, SAE L2; SAE J3016, 2018) or even merely being ready to act as a fallback level in case the automation reaches its limits (so-called “conditionally automated driving”, SAE L3). Consequently, an automated driving system’s (ADS) HMI should effectively support the driver in interactions with the ADS, such as handing control over to the automation (Borojeni et al., 2017), supervising the automation and driving environment (van den Beukel & van der Voort, 2017) or resuming vehicle control when necessary (Naujoks, Mai, & Neukum, 2014). In this context, different interaction concepts and interfaces have been proposed in research papers and in series production vehicles (e.g., Manca, De Winter, & Happee, 2015; Mirmig et al., 2017).

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Most of these interfaces share a variety of elements that are intended to keep the driver informed about her/his current tasks as well as the status, actions and intentions of the automation (Manca, De Winter, & Happee, 2015), such as pictorial mode indicators (e.g., automation off/available/on, Naujoks, Purucker, Neukum, Wolter, & Steiger, 2015; Naujoks, Purucker, et al., 2017), indicators showing the reliability level of the automation (Beller, Heesen, & Vollrath, 2013; Helldin, Falkman, Riveiro, & Davidsson, 2013; Hergeth, Lorenz, & Krems, 2017) and text boxes to deliver messages to the driver (Manca et al., 2015; Naujoks, Forster, Wiedemann, & Neukum, 2016). Increasing automation transparency can promote trust and acceptance as well as safety of control transitions to and from automated mode (Forster, Naujoks, & Neukum, 2017; Forster, Naujoks, Neukum, & Huestegge, 2017; Naujoks, Forster, Wiedemann, & Neukum, 2017). However, these HMI design approaches will only support the safe and efficient use of ADS if ergonomic principles of human-machine interaction are taken into account. For example, cluttered and unnecessarily complex interfaces might lead to distraction rather than efficient communication (Pankok & Kaber, 2017). Legibility issues might lead to high visual (Reimer et al., 2014; Sawyer, Dobres, Chahine, & Reimer, 2017) and mental workload (Pauzié, 2008) when information from the HMI is retrieved. Displays positioned out of the driver's expected field of view might fail to attract attention when needed (Lamble, Laakso, & Summala, 1999; Vallerie, 1967; Wittmann et al., 2006).

Therefore, regulatory bodies have already started to ask manufacturers and other entities to verify that the interfaces of automated vehicles are capable of enabling the driver to safely interact with them. For example, the National Highway Traffic Safety Administration (NHTSA, 2016) has been the first federal agency to demand that indicators should be capable of informing the driver of different automation modes, such as that the automation is working reliably, that it is engaged in an automated driving mode, that it is not available for automated driving, or even requesting a control transition. Apart from studies involving human subjects, which can be both time-consuming and cost-intensive, a basic verification of these demands could be accomplished by investigating whether the interface is in compliance with applicable and commonly agreed standards of automotive HMI design, such as standards and guidelines released by NGOs, research institutes and associations (e.g., Alliance of Automobile Manufacturers (Auto Alliance), Japanese Association of Automobile Manufacturers (JAMA), etc.). Another advantage of such an analytical approach is its high flexibility and the fact that it could be used to guide HMI development, thereby preventing non-compliance at early stages of the product development cycle. Comparable work has already been conducted in the area of in-vehicle information systems (IVIS) that can be used during manual vehicle control (Stevens, Quimby, Board, Kersloot, & Burns, 2002). As there are no standards that apply specifically to HMIs for automated driving, a review of available documents was conducted and compiled into a checklist. The focus of this checklist is on visual-auditory and visual-vibrotactile interfaces, as these are most commonly used in series production vehicles and in research studies (see Mirnig et al., 2017, for a recent review).

In this paper we offer an initial set of principles, criteria, and verification methods for guiding the development of automated vehicle HMIs which reflects the work of the authoring research team. As such, it should be viewed as a starting point that may still contain gaps where research in the field of automated vehicles is still emerging. It expresses *one set* of guidelines and is intended as a *first step* towards guideline development and vehicle-verification methods. They are shared through this publication as a means of (1) stimulating a discussion within the larger scientific and technical community towards future standards that may be appropriate for guiding the development of automated vehicle HMIs and (2) facilitating a more rapid convergence within the technical community towards an agreed-upon set of robust guidelines and verification methods that can serve the industry in the important evolution of automation within vehicles.

2. Applicable HMI standards, guidelines and best practices

Today, in-vehicle information systems (IVIS) such as navigation systems or in-car entertainment systems are available in almost every passenger vehicle. The visual channel is the predominant output modality, relying on displays mounted to the head unit or the instrument cluster for information presentation (Meixner et al., 2017). Using visual-manual IVIS during driving inherently causes glances away from the road (Purucker, Naujoks, Prill, & Neukum, 2017), which elevates the risk of being involved in safety-critical situations when these tasks are too complex (e.g., requiring several operating steps or glances away from the road, Klauer, Dingus, Neale, Sudweeks, & Ramsey, 2006; Klauer et al., 2014). However, mental workload caused by IVIS tasks can also impair driving performance (e.g., reaction times to road hazards) even when these tasks are primarily voice-based (Alm & Nilsson, 1995; Mehler et al., 2016; Patten, Kircher, Östlund, & Nilsson, 2004; Reimer, Mehler, Reagan, Kidd, & Dobres, 2016).

To counteract IVIS-induced impairments of driving performance, several organizations such as the Society of Automotive Engineers International (SAE), International Standardisation Organization (ISO), the Alliance of Automobile Manufacturers (Auto Alliance) or the National Highway Traffic Safety Administration (NHTSA) have published guidelines and standards to guide the design process of Human-Machine Interfaces. Prominently used guidelines are also available for the design of advanced traveller information systems (ATIS, Campbell, Carney & Kantowitz, 1997) and driver-vehicle interfaces in general (Campbell et al., 2016). These guidelines touch several aspects that are also relevant to the HMI design of automated vehicles, such as the positioning of interfaces, colour coding or auditory output characteristics.

Apart from IVIS, guidelines and standards concerning the design of in-vehicle warnings can support HMI design for automated vehicles with regard to mandatory control transitions from automated to manual driving. In these situations, the driver is expected to resume manual control within a limited time frame (by so-called “take-over requests” (TOR) or “requests

to intervene” (Rtl), [Marberger et al., 2017](#)). As in the case of in-vehicle warnings, the urgency of the situation and the required actions need to be communicated quickly and unambiguously.

In the next section, the aforementioned sources (i.e., applicable standards, guidelines and recommendations for the design and assessment of IVIS, ATIS, driver-vehicle interfaces and in-vehicle warning systems) are presented in more detail and compiled into guidelines for the design and assessment of automated vehicle HMIs. The sources (standards, guidelines, best practices and empirical studies) were selected by the authoring research team to reflect the requirement that HMIs for ADS should be designed in accordance with relevant best practices and design principles ([NHTSA, 2016](#)) to facilitate the usability of ADS (i.e., that the driver can effectively, efficiently and satisfactory use the ADS; [ISO 9241, 2018](#)). In this sense, the scope of these guidelines is twofold:

- First, they are intended to ensure that mode indicators of the ADS needed for a safe and comfortable use (including control transitions) are present in the HMI and non-intended activation and deactivation of the vehicle automation is prevented.
- Second, from an information processing perspective (see [Wickens, 1984](#)), the guidelines focus on supporting the users of vehicle automation in *noticing* (e.g., by placing visual interfaces close to the driver's expected line of sight) and *interpreting* these messages (e.g., by intuitive colour coding and the use of generally accepted symbols), which is a prerequisite for the mode indicators having their intended effect on the user. From the user's perspective, overlooking mode changes or misinterpreting mode indicators can have serious consequences.

The guidelines are supported by providing relevant research findings and best-practices on how to fulfill the requirements. The guidelines are then transferred into a simpler checklist that is presented in the subsequent chapter. It should be emphasized that the scope of the HMI guidelines is to facilitate regular usage of the ADS, such as switching the automation on, checking the status of the ADS or taking back manual control from the automation (either driver-initiated or as a result of a TOR). A more comprehensive evaluation of the ADS will most likely need to incorporate (a) usability testing with real users in instrumented vehicles or high-fidelity driving simulators and (b) investigating other domains such as the assessment of *controllability* of system limits or failures and foreseeable *misuse* of the ADS as defined in the RESPONSE Code of Practice ([RESPONSE Consortium, 2006](#)). The aim of the proposed method is thus not to replace usability testing with participant samples, but to complement empirical approaches with an economical heuristic evaluation tool. Limitations of the current approach are also discussed in the corresponding sections of this paper.

3. Guidelines for automated vehicles HMIs

3.1. Operation principles

Guideline #1: Unintentional activation and deactivation should be prevented.

Activation and deactivation of the automated driving functionality transfers control over the vehicle between the human operator and the automation. As research has shown that drivers usually need several seconds to safely come back into the control loop ([Salvucci, Taatgen, & Borst, 2009](#)), it must be guaranteed that they do not accidentally deactivate the automation. One possibility could be to require that drivers put both hands back on the steering wheel ([AdaptiVe Consortium, 2017](#)) before the system can be deactivated. However, it should also be ensured that drivers can easily deactivate the ADS when they are requested to do so or intend doing so by themselves. This could be ensured by self-explanatory labelling of system controls (see below for more information) and by positioning control elements where they are easily reachable and do not afford taking both hands off the steering wheel.

On the other hand, non-intentional activation of the automation needs to be prevented as well. For L2-automation, the driver must ensure operational readiness before automated driving features can engage ([CAMP, 2016](#)). It is also recommended that L3-automation is not active at the beginning of each drive, but that it can automatically resume once activated within the drive when appropriate situational circumstances for safe automated operation are met ([CAMP, 2016](#)).

3.2. Indication of system mode

Guideline #2: The system mode should be displayed continuously.

Different driving automation levels correspond with different responsibilities and possibilities of the driver, such as the need for supervision (SAE L2) or the possibility to engage in non-driving-related activities (SAE L3). Therefore, the user should be continuously informed about the system mode so that she/he is supported in keeping a sufficient level of mode awareness ([CAMP, 2016](#); [TRL, 2002](#)). According to the NHTSA, this requires a minimum set of distinguishable mode indicators that should be present in the HMI, informing the user that the system is:

- (1) functioning properly
- (2) currently engaged in an automated driving mode

- (3) currently unavailable for automated driving
- (4) experiencing a malfunction
- (5) requesting a control transition from the automated driving system to the operator (NHTSA, 2016).

When there are several automated driving modes (such as L2 and L3-mode) within the same system, these indicators should be present separately for all sub-systems. Other indicators, such as driver re-engagement messages (e.g., request to monitor the road or put the hands back on the steering wheel) may also be part of the HMI and should also be evaluated with the following guidelines.

Guideline #3: System state changes should be effectively communicated.

The communication of mode changes is of primary concern for a safe and efficient HMI for automated vehicles (Adaptive Consortium, 2017). For example, missing the disengagement of an L2 automation might end up in safety-critical situations or even accidents. Therefore, transitions between system modes should be reflected in the HMI (i.e., there should be a change of a pictorial indicator or an auditory message). Here, it is of utmost importance that changes in the driver's responsibility are made clear (e.g., by using a disclaimer that the driver has to monitor the automated driving system and the vehicle environment in case of a transition to L2, or that the driver still has to be available as a fallback level after a transition to L3). However, it should be noted that significantly more research is needed on how to achieve this guideline. Although there is a growing body of research on this topic, the best way of achieving the goal of effectively communicating the driver's responsibility as a change of automation modes is still to be determined.

When there are notable time lags between input and system reaction, the system should provide feedback about the progress of a state transition. This includes displaying that the system state requested by the user is not yet in effect (e.g., when an L2 system has been activated, but the lateral control only becomes active after certain conditions are met). Effectively communicating mode changes may also necessitate displaying whether a user-initiated control transition has failed (e.g., when the user tries to activate an ADS outside of its operational design domain or other pre-conditions necessary for system activation are not fulfilled). Again, efficiently communicating the progress (or failure) of the activation or deactivation of the ADS is still an area in which more research is needed, as no standard or best practice to fulfill this HMI guideline is currently available.

3.3. Display installation and information presentation

As the most common method used in the automotive domain to communicate information to the driver still is the visual channel, the following guidelines are intended to facilitate the design of mode indicators as well as mode transitions by ensuring that they can be easily perceived, distinguished from one another and correctly interpreted by the human operator of the automated vehicle.

Guideline #4: Visual interfaces used to communicate system states should be located at a suitable position and distance. High-priority information should be presented close to the driver's expected line of sight.

When the driver is not required to pay (continuous) attention to the road scene during automated driving, in-vehicle displays used for automated vehicles could possibly be mounted in different positions than the ones optimized for manual or assisted driving. However, it should be kept in mind that it will probably require the driver more time to access information the farther these displays are moved away from her/his line of sight (O'Regan & Levy-Schoen, 2013). In the worst case, onset of important information may be missed altogether. That is because the human visual system is insensitive to visual stimuli in the periphery. For example, colours (Vallerie, 1967; Young, 2003) are more difficult to distinguish when presented in the periphery. Perceived blink rate of stimuli, which is often used to convey urgency, also decreases when stimuli are presented in the periphery (Jacob & Cam, 2003).

Therefore, it is recommended to position visual interfaces within a 30-degree cone about the user's normal line of sight (JAMA, 2004; MIL-STD-1472G, 2012; TRL, 2002). A further application and verification procedure of the requirement that the downward viewing angle should be less than or equal to 30° at the geometric centre of display can be found in the Auto Alliance Guidelines (Auto Alliance, 2006; principle 1.4). Ideally, safety critical information such as TORs should be presented in a 20-degree cone (HARDIE Consortium, 1996; SAE J2400, 2003).

Building on previous research on manual and assisted driving, Head-Up Displays (HUD) may provide a promising way of increasing the probability that mode changes are noticed and easily processed by the driver. In general, HUDs go along with reduced information processing costs and increased eyes-on road times (Kiefer et al., 1991; Kiefer, 1998a, 1998b). For example, it has been demonstrated that drivers recognize the onset of telltales with a higher probability when using HUDs compared to Head-Down Displays (HDDs; Grant, Kiefer, & Wierwille, 1995), and that they are better able to react to appearing road hazards when they access information from the HUD (Kiefer & Gellatly, 1996; Kiefer, 1998a, 1998b). HUDs may also provide the opportunity to present information that cannot be effectively communicated through HDDs (Kiefer, 1996). Given these findings, there is ample evidence that HUDs can be used beneficially to communicate mode changes in the context of ADS. However, when using ADS with L3-capability, it is expected that the drivers will perform some sort of non-driving related tasks (SAE, 2018), freeing her/him from continuous monitoring of the driving environment. In such cases, presenting information related to the HMI on the same display used for a non-driving related task can assure that guideline #4 is met.

Another way of avoiding the drawbacks of peripheral displays could be using large peripheral interfaces or peripheral movement to attract the driver's attention (Dzennius, Kelsch & Schieben, 2016).

Guideline #5: HMI elements should be grouped together according to their function to support the perception of mode indicators.

Another reason to carefully position relevant information to the driver during automated driving is that any glance transition necessary to process the information will inherently prolong reaction times, increase visual workload and promote perception errors. Therefore, visual elements used to convey relevant state transitions of the automation should be grouped together (AdaptiVe Consortium, 2017; NASA-STD-3001, 2011; TRL, 2011) whenever this avoids unnecessary glances to different regions of visual displays. An example may be a pictorial indicator of a systems state that is supplemented by a text explanation (e.g., an icon representing L2-functionality together with a text explanation to “monitor the road”). Displaying such elements too far away from each other will increase reading times and the probability of missing parts of the information altogether. It is also recommended that the display arrangement allows high-priority messages (e.g., TORs, disengagement of the L2-functionality, etc.) to be easily distinguishable from low-priority information (e.g., appearing in their own dialog box or pop-up; JAMA, 2004; MIL-STD-1472G, 2012).

Guideline #6: Time-critical interactions with the system should not require continuous attention.

The driver will be more or less vulnerable while interacting with the ADS in different driving situations. For example, when activating the system, the driver is still fully responsible for the driving task until the driving automation is active. The same goes for system deactivations when control is transferred back to the user of the ADS. In these situations, the driver is performing the dynamic driving task and possibly needs to react to potentially critical driving situations. Here, information presented in the HMI should not necessitate that the driver is gazing at the respective displays continuously (JAMA, 2004).

3.4. Legibility

Guideline #7: The visual interface should have a sufficient contrast in luminance and/or colour between foreground and background.

When information is retrieved by the driver, extraction of visual information from the HMI should be possible “at a glance”, especially in those interaction sequences in which the driver still has to divide attention between monitoring the road and the HMI (such as activating the automation or supervising an L2-automation). The ease with which a character or a symbol on the display is identified can be understood as its legibility (Humar & Gradis, 2008). Poor legibility can increase reading times and thus decrease the efficiency of information processing. Up to some asymptotic level, legibility can be increased by luminance and colour contrast (Wu & Yuan, 2003 Legge, Parish, Luebker, Wurm, 1990). Thus, the colour combinations used in the visual interface to display system states will have a great impact on its legibility (Humar & Gradis, 2008). In the automotive domain, the predominant combinations are that of coloured or white texts/icons on black background. Certain colour combinations, such as that of blue on black background can lead to low legibility and reduced reading accuracy and should therefore be avoided (see Table 1). In general, a contrast ratio of foreground and background between 3:1 should be used, but a higher contrast is recommended (HARDIE Consortium, 1996; TRL, 2002); especially in night mode, a contrast ratio of 5:1 is recommended (ISO 15008, 2017).

Guideline #8: Texts (e.g., font types and size of characters) and symbols should be easily readable from the permitted seating position.

Table 1

Subset of recommended colour combinations according to ISO 15008. – = not recommended, 0 = acceptable, + = good, ++ = very good. The table only shows the background colours white and black; other background colours can be found in ISO 15008.

Colour of character or symbol	Background colour	
	White	Black
White		++
Yellow	–	++
Orange	0	+
Red, purple	+	+
Green, cyan	+	+
Blue, violet	++	–
Black	++	

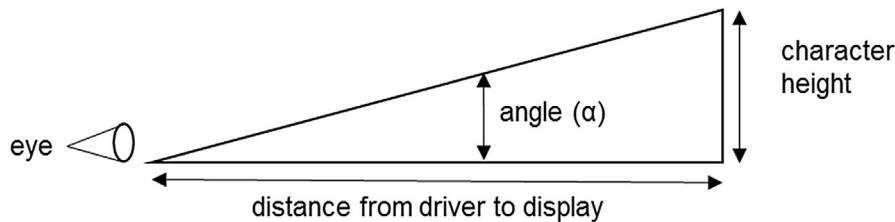


Fig. 1. Schematic for estimation of character height. (adapted from TRL, 2002).

However, using an adequate contrast is not a guarantee for sufficient legibility. Size as well as inter/intra-target spacing also plays an important role (Sanders & McCormick, 1993). Most evidently, small typefaces and low display resolutions go along with increased reading times and reading errors in digital displays (Tullis, Boynton & Hersh, 1995; Huang, Rau & Liu, 2009). Therefore, the commonly accepted minimum required character height is a visual angle of 12', 16' is considered as acceptable, but at least 20' is recommended (ISO 15008, 2017). However, it should be noted that earlier guidelines required a bigger minimum height of 15' (TRL, 2002), and recommended a greater height of 24' (TRL, 2002) (see Fig. 1).

Text reading is impeded by hard to read text-fonts. Sans-Serif fonts are recommended (e.g., Arial or Helvetica; Green, Levison, Paelke, & Serafin, 1994; HARDIE Consortium, 1996; MIL-STD-1472G, 2012; TRL, 2002), especially for isolated words and short messages. The ratio between the stroke width and character height should be between 10% and 20% (ISO 15008); the more critical the information, the wider the stroke should be (see also Green et al., 1994; TRL, 2002). The relation between the width and the height of the character should be between 65% and 80% (ISO 15008). The more critical the information, the wider the character should be (MIL-STD-1472G, 2012; TRL, 2002) (see Fig. 2).

3.5. Understandability

Guidelines 2–8 had the aim of maximising the probability that a relevant mode transition presented via the HMI of an automated driving system can be noticed (e.g., by positioning of displays) and processed by the user with minimal effort (e.g., through adequate colour contrast and character presentation). However, apart from the mere registration of a message, the driver will need to derive the right consequences from it (e.g., monitoring the road, putting the hands back on the steering wheel, taking over control, etc.). At least in the introductory period of automated vehicles, the challenge of creating intuitive messages should not be underestimated. For example, surveys show that the general public does not easily understand the roles and responsibilities associated with commonly used labels for automated driving features, such as “pilot” or “assist” (Abraham, Seppelt, Mehler, & Reimer, 2017). The design of unambiguous HMI messages should therefore be a primary concern that can be supported by the following guidelines.

Guideline #9: Commonly accepted or standardized symbols should be used to communicate the automation mode. Use of non-standard symbols should be supplemented by additional text explanations or vocal phrases.

Automated driving features will bring new complexity to interacting with assisted and automated driving systems. For example, according to the operational design domain (ODD), an automated driving system may have L3-capability or L2-capability during the same ride, with the latter possibly varying in the degree of required driver engagement (e.g., having hands on the steering wheel and eyes on the road or not). On the visual HMI, the respective system modes will most likely be represented by a primary symbol such as a tell-tale or a pictogram. To ensure safe and efficient use, it must be made sure that these are easily understood. For example, requesting the driver to put the hands back on the steering wheel (as part of a driver monitoring strategy in SAE L2) and requesting the driver to take over control (as part of a take-over request in SAE L3) has often been implemented by using very similar symbols of hands on a steering wheel, although they have drastically different consequences. To the naïve system user, these symbols should be made as intuitively meaningful as possible by using commonly accepted or standardized symbols (HARDIE Consortium, 1996; TRL, 2002), however, it should also be pointed out that a commonly agreed-upon pictorial set of mode indicators is not yet available. The use of non-standard symbols should

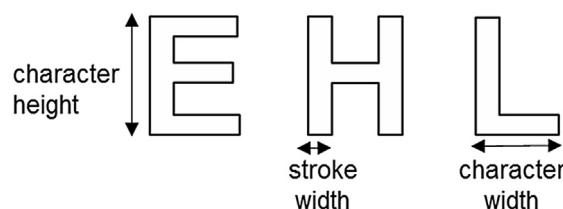


Fig. 2. Schematic for estimation of ratio of character height to width and stroke width. (adapted from TRL, 2002).

be supplemented by additional text (TRL, 2002), possibly using a designated text box area (HAVEit Consortium, 2011) or an equivalent voice message. In this context, voice messages might unburden the driver's visual processing channel by allowing to process HMI messages without the need to take one's eyes off the road. However, it should be noted that in view of users with hearing impairments, voice messages might help most but not all users to understand non-familiar symbols used to communicate AD modes.

Guideline #10: The semantic of a message should be in accordance with its urgency.

While interacting with an automated vehicle, HMI messages will vary in their urgency – for example, regarding the need for manual control. In this regard, a request to put the hands back on the steering wheel or to look on the road as part of a driver monitoring strategy in the absence of any critical driving event has not the same urgency as an imminent take-over request. Scheduled take-over requests that display an upcoming system limit well in advance are also to be distinguished from imminent take-over requests. It has been proposed to present non-critical information in an informative style (e.g., “end of automated drive approaching”), but to display safety-critical information in a command style (e.g., “take-over driving immediately!”; TRL, 2002). The wording itself is also crucial: For example, the word “danger” will evoke higher perceived urgency than “warning” or “caution” (Bazilinsky and de Winter, 2015). Thus, it should be ensured that the semantic is in accordance with the intent of the HMI indicator, supporting the driver in a timely reaction.

Guideline #11: Messages should be conveyed using the language of the users (e.g., national language, avoidance of technical language, use of common syntax).

Besides the urgency of a message, it must be assured that its meaning is clearly understood. This point is especially important in systems with changing automation levels in which the driver must be explicitly informed when she/he has to monitor the situation or even take over manual driving (AdaptIVe Consortium, 2017). For example, it has been shown that the word “assist” can create confusion whether the system is “assisting” the driver or whether the system “requires assistance” from the driver (Abraham et al., 2017). Likewise, technical terms that describe automated driving systems with different capability, such as the ones developed by SAE, NHTSA or BAST, may not be intuitively understood by users. From a general perspective, the visual display should speak “the language of the user” (Auto Alliance, 2006) so that it supports timely and intended reactions (UNECE, 2009). The following recommendations should therefore be considered: (1) use of national language (HARDIE Consortium, 1996; TRL, 2002), (2) use of simple and non-technical language (Green et al., 1994; HARDIE Consortium, 1996; TRL, 2002), (3) avoidance of abbreviations (HARDIE Consortium, 1996; TRL, 2002) and (4) displaying the functionality of the automation rather than the SAE/NHTSA/BAST-level (AdaptIVe Consortium, 2017).

Guideline #12: Text messages should be as short as possible but as long as necessary.

While displayed text should be as intuitive as possible, excessively long messages should also be avoided (AdaptIVe Consortium, 2017; Campbell et al., 1997; SAE J2831, 2012; TRL, 2011) as these likely lead to overly complex displays and increase reading times. It is thus recommended to display not more than four units of information in one HMI message (HARDIE Consortium, 1996; TRL, 2002).

3.6. Colour coding

Colour coding is an efficient way to support that drivers notice mode changes as well as the consequences associated with them. While guidelines 9–12 focused on possibilities of coding mode-related information verbally, guidelines 13–17 are aimed at optimizing non-verbal information presentation. For example, using the colour red will draw the driver's attention to a display and communicate the high urgency of the situation, while a green colour will naturally communicate that a system is working error-free. However, as colour codes need to be decoded by the user, there is also a fair possibility of misinterpretation. Thus, efficient and unambiguous colour-coding should be of primary concern for the design of efficient HMIs for automated vehicles.

Guideline #13: Not more than five colours should be consistently used to code system states (excluding white and black).

On a basic level, the colours used to code automated driving modes should be used consistently (e.g., indicators pertaining to a driving mode should resemble one another, AdaptIVe Consortium, 2017). To support the driver in learning and recognizing different system modes, it is generally recommended not to use more than five different colours (HARDIE Consortium, 1996; TRL, 2002, 2011).

Guideline #14: The colours used to communicate system states should be in accordance with common conventions and stereotypes.

As colour is a natural, non-verbal way of communicating with the user of the automated driving system, it should be assured that the colours associated with different system modes are in accordance with common stereotypes (Campbell, Richard, Brown, & McCallum, 2007; Green et al., 1994; HARDIE Consortium, 1996; MIL-STD-1472G, 2012; TRL, 2002). For example, red usually communicates an imminent danger, yellow/amber a state of alertness and green a hazard-free operating state (Campbell et al., 2007; Green et al., 1994; HARDIE Consortium, 1996; TRL, 2002).

Guideline #15: Design for colour-blindness by redundant coding and avoidance of red/green and blue/yellow combinations.

Colour blindness should also be accounted for in the display. For example, about 9% of the male population has a green/red colour deficiency, whereas yellow/blue colour deficiencies are somewhat lower. Differentiating automated driving states will be nearly impossible for them when (1) combinations of these colours in the same display element are not avoided (e.g., red hands displayed on a green display) or (2) system states are not redundantly coded in a suitable way (HARDIE Consortium, 1996; TRL, 2002, 2011).

3.7. Auditory and vibrotactile messages

As the driver's visual channel is already loaded because she/he is observing the traffic environment (up to SAE L2 automation) or probably engaged in some sort of non-driving related task (SAE L3 automation and higher), multimodal HMIs are usually used to draw the driver's attention to the HMI or relevant events in the traffic environment. Auditory and vibrotactile messages have been successfully applied in contexts such as forward collision warnings. The use of more complex auditory signals, such as speech output or earcons, might even make it unnecessary to retrieve any information from visual displays at all. Auditory and vibrotactile messages should be designed in accordance with the following guidelines to make them as efficient as possible.

Guideline #16: Auditory output should raise the attention of the driver without startling her/him or causing pain.

Generic auditory output usually consists of a series of signal bursts that vary in length, loudness and frequency to convey urgency. First of all, signal bursts should be long enough to be noticeable by the driver. Signal durations between 100 and 500 ms are recommended (Green et al., 1994; HARDIE Consortium, 1996). Second, auditory outputs presented to the driver should be loud enough to be recognizable, however, they should not be startling or causing pain. In general, it is recommended to use a loudness between 50 and 90 dB (TRL, 2002) and to avoid using a loudness greater than 120 dB (Green et al., 1994; HARDIE Consortium, 1996). Considering that there will be some background noise while the vehicle is in motion, the outputs should be between 15 dB and 30 dB above the background noise (Green et al., 1994; HARDIE Consortium, 1996; TRL, 2002, 2011). Frequencies between 500 and 4000 Hz are recommended (Green et al., 1994; HARDIE Consortium, 1996).

Vibrotactile output also consists of signal bursts with a defined length, frequency and strength. Comfortable stimuli range 15–20 dB above the detection threshold for the respective body region (ETSI EG 202 048, 2002; Van Erp, 2002). Amplitudes above 0.6–0.8 mm should be avoided as they will likely result in a pain sensation (Van Erp, 2002). Regarding signal frequency, optimal sensitivity is achieved at frequencies between 150 and 300 Hz for all body sites; at lower and higher frequencies, the displacement of the skin must be greater to be detected (Choi & Kuchenbecker, 2013; Jones & Sarter, 2008). Signal bursts should be between 50 ms and 200 ms, as longer vibrations are usually perceived as annoying (Choi & Kuchenbecker, 2013; Jones & Sarter, 2008; Petermeijer, De Winter, & Bengler, 2016).

Guideline #17: Auditory and vibrotactile output should be adapted to the urgency of the message.

To make sure that drivers respond quickly to urgent messages (e.g., imminent TORs), it is a necessity that the intrusiveness of messages is adapted to their urgency (Adaptive Consortium, 2017; HARDIE Consortium, 1996). Presenting different messages always with the same urgency might cause alarm fatigue and ultimately slow down reaction times when fast reactions are needed (Naujoks, Kiesel & Neukum, 2016). Therefore, imminent, cautionary and notification messages should be easily distinguishable from each other (ISO 15008, 2017; ISO 15623, 2013). Ways of conveying urgency for generic warnings are increasing the pulse rate (Bazilinsky and de Winter, 2015; ISO 15623, 2013), the frequency (HARDIE Consortium, 1996; ISO 15623, 2013) or the loudness of the output. When no immediate attention of the driver is needed, either unobtrusive (ISO 15623, 2013) or no auditory output should be used (Green et al., 1994).

Regarding vibrotactile stimuli, it is recommended to code urgency through vibration intensity (ETSI EG 202 048, 2002; Meng & Spence, 2015), size of vibration area (ETSI EG 202 048, 2002), decreased inter-pulse-interval, pulse duration, pulse repetition rate or number of pulses (Petermeijer et al., 2016). To code different information, a variation of the location and timing dimensions rather than that of the frequency and amplitude is recommended (Meng & Spence, 2015).

3.8. Warning messages

Guideline #18: High-priority messages should be multimodal.

High-priority messages should be presented in more than one modality for several reasons. First, this usually speeds up processing time by the so-called redundancy gain. Thus, visual-auditory or visual-vibrotactile warnings have been shown to lead to faster responses than unimodal warnings (Campbell et al., 2007). Second, auditory or vibrotactile signals are a useful way of delivering information, but they are also short-lived in the sense that information can be missed or forgotten. This is especially important in case of critical information such as imminent TORs. To reduce the need for serial information processing and retaining auditory/vibrotactile information in working memory, it is thus recommended that high-priority auditory/vibrotactile information is also presented visually (HARDIE Consortium, 1996; TRL, 2002, 2011).

Guideline #19: Warning messages should orient the user towards the source of danger.

Control transitions from automated to manual driving might require the driver to manually guide the vehicle through potentially critical situations the automated vehicle cannot handle. In this case, imminent messages should induce an orienting response, causing the driver to look in the direction of the imminent hazard when appropriate (e.g., from front/side/back of the vehicle; [HARDIE Consortium, 1996](#); [SAE J2400, 2003](#)). In case manual control is needed, focusing the driver's attention on the display should be avoided ([JAMA, 2004](#)).

Guideline #20: In case of sensor failures, their consequences and required operator steps should be displayed.

Sensor failures might cause an automated driving system, or an associated sub-system, to become unavailable for use. An example might be an optical sensor that is covered by mud or snow, which might trigger an ADS to become unavailable ([NHTSA, 2016](#)). In this case, the consequences of the current status or system malfunction, particular on vehicle control and manoeuvring, as well as required operator steps, should be displayed ([SAE J2400, 2003](#); [UNECE, 2009](#)).

4. Guideline application and verification procedure

4.1. Introduction

[Table 2](#) summarizes the guidelines presented in the last section. These can be applied in several ways in a product development context. First, they can be used as a reference by designers and engineers to guide the development of automated vehicle HMIs in an early conceptual development stage. Second, they can be used as an efficient tool to verify whether the HMI of a prototype or series production vehicle is complying with the guidelines and is thus in accordance with international standards and best practices as presented in this paper. For this purpose, a checklist has been developed that will be presented in this section.

4.2. Scope of the heuristic assessment

Evaluation methods used to assess user interfaces can either be formal or informal. While formal methods use exact models or formulas to calculate usability measures, informal methods typically consist of a heuristic evaluation methodology ([Nielsen, 1994](#)) that is based on inspection of the HMI by expert evaluators examining the interface and judging its compliance with recognized usability principles or guidelines (the so-called “heuristics”) while performing typical user tasks. The evaluators identify usability problems or concerns, record them (together with their severity) and link them to the heuristic design principles that have not been met. When properly conducted, such heuristic assessments can even be more successful in identifying usability issues than a usability test with participants ([Desurvire, Kondziela, & Atwood, 1992](#)). Usually, it is possible to identify most usability problems with small samples consisting of four to five trained expert evaluators ([Jeffries, Miller, Wharton & Uyeda, 1991](#); [Virzi, 1992](#)). The results of the heuristic evaluation can then be used to guide further improvements in the HMI that can help resolve each identified usability problem. We propose an adaptation of this

Table 2
Checklist items.

#	Item
1	Unintentional activation and deactivation should be prevented
2	The system mode should be displayed continuously
3	System state changes should be effectively communicated
4	Visual interfaces used to communicate system states should be mounted to a suitable position and distance. High-priority information should be presented close to the driver's expected line of sight
5	HMI elements should be grouped together according to their function to support the perception of mode indicators
6	Time-critical interactions with the system should not afford continuous attention
7	The visual interface should have a sufficient contrast in luminance and/or colour between foreground and background
8	Texts (e.g., font types and size of characters) and symbols should be easily readable from the permitted seating position
9	Commonly accepted or standardized symbols should be used to communicate the automation mode. Use of non-standard symbols should be supplemented by additional text explanations or vocal phrase/s
10	The semantic of a message should be in accordance with its urgency
11	Messages should be conveyed using the language of the users (e.g., national language, avoidance of technical language, use of common syntax)
12	Text messages should be as short as possible
13	Not more than five colours should be consistently used to code system states (excluding white and black)
14	The colours used to communicate system states should be in accordance with common conventions and stereotypes
15	Design for colour-blindness by redundant coding and avoidance of red/green and blue/yellow combinations
16	Auditory output should raise the attention of the driver without startling her/him or causing pain
17	Auditory and vibrotactile output should be adapted to the urgency of the message
18	High-priority messages should be multimodal
19	Warning messages should orient the user towards the source of danger
20	In case of sensor failures, their consequences and required operator steps should be displayed

methodology based on the formulated guidelines to achieve a holistic, but also efficient evaluation and verification of ADS HMIs.

As efficiency is extremely important in the product development context, and time and monetary resources may not always allow extensive user testing, a comprehensive and thoroughly itemized checklist evaluation can be used to verify that basic HMI requirements are met. When sufficient resources are available, however, the checklist evaluation can be conducted separately and complementary to an empirical usability test with real users in which objective and self-report data from users interacting with the ADS are collected and analyzed from an on-road instrumented vehicle or a realistic high-fidelity driving simulator. It should be emphasized that a truly comprehensive usability assessment of user interfaces should always rely on both empirical testing methods and expert assessments such as the one proposed in this paper as they tend to uncover different types of usability issues (Jeffries & Desurvire, 1992).

4.3. Proposed heuristic evaluation methodology

To verify compliance with the proposed guidelines, it is recommended that the HMI of an ADS should be evaluated by four to five independent raters in an on-road test (Nielsen & Molich, 1990). We propose the test procedure to involve the following steps:

- First, a set of test cases should be defined. This will require gathering a clear understanding of the system modes and transitions within an ADS. Depending on the focus of the evaluation as well as practical constraints (e.g., time or budget), different use cases might be selected. An example could be testing the activation and deactivation procedure, as well as take-over requests. A systematic approach to the selection of use cases for the assessment of L3 systems was recently proposed by the authoring team (Naujoks, Hergeth, Wiedemann, Schömig & Keinath, 2018)
- Depending on the test cases, a test plan should be developed that allows different evaluators to produce these test cases (e.g., taking the hands off the steering wheel to produce a hands-off warning, driving through a work zone to produce a disengagement of lateral control, etc.). Both the test cases and the method of producing the test cases should be documented.
- The test cases should then be completed by at least four to five evaluators during an on-road assessment (Jeffries et al., 1991; Virzi et al., 1992). The external conditions of the on-road test should be documented (e.g., time of day, light conditions, traffic volume). The purpose of this documentation is that the test should be conducted under “everyday” usage situations (i.e., the results should not be attributable to extreme conditions) and to make them comparable to other test sites.
- To ensure that the results are not based on individual opinions, the initial results should be compared between the raters by calculating inter-rater reliability scores (Hallgren, 2012). When the ratings do not sufficiently converge, it is recommended to reject the results. The results of the rater agreement check should be documented.
- If rater agreement has been demonstrated, checklist items with diverging agreement should be discussed to come to a final, unanimous assessment. The consolidated result should be documented as well. Special emphasis should be given on checklist items in which the rater team unanimously attests a deviation from the guidelines.

To support the verification test, an extended checklist of the guidelines and a reporting sheet can be found in Appendix A.

5. Discussion

5.1. Summary

In this paper we compiled an initial set of guidelines for the design of automated vehicle HMIs that are intended to verify compliance with basic requirements formulated in applicable international standards and guidelines for the design of HMIs. It is intended as a first step towards further guideline development and is aimed at stimulating discussion within the larger scientific and technical community towards future standards, and to facilitate more rapid convergence within the technical community towards an agreed-upon set of robust guidelines and verification methods for automated vehicles HMIs. Relevant research findings were collected to compliment the guidelines and to serve as a guidance for an expert-based usability assessment which can be used in the process of HMI development and verification. For the latter, a heuristic evaluation procedure on how to demonstrate compliance with the guidelines was also proposed.

5.2. Limitations

It should be noted that the focus of this paper was to present a set of most relevant guidelines that can be used for a basic verification of the HMI by usability experts. For the sake of practicability and efficiency, these were kept as short as possible, however, it is likely that they cannot cover every aspect of the HMI for ADS. At this point it is important to emphasize that new and innovative HMI designs may rely on other HMI elements as the ones covered by the sources used to compile the checklist. In this case, the raters are encouraged to give a positive assessment when they judge the respective guideline to be fulfilled during the on-road test, even if this cannot be based on the design recommendations. In case of sufficient inter-rater agreement, verification of the guideline can be assumed.

Furthermore, other relevant evaluation criteria of HMIs for automated vehicles are out of the scope of the proposed method, such as the *controllability* of system limits and failures. Apart from the automated vehicle's HMI, the reliability and limits of the automated driving system will greatly impact the safety of use of the system (Gold, Naujoks, Radlmayr, Bellem, & Jarosch, 2017; Marberger et al., 2017). In particular, infrequent and unexpected disengagements might be difficult to handle by the human operator (Parasuraman and Riley, 1997). Likewise, providing not enough time for re-engagement in the driving task will leave the user unprepared to cope with the demands of the driving situation (Gold, Damböck, Lorenz & Bengler, 2013). Evaluation of these HMI aspects enables empirical approaches to be taken, based on a precise specification of situational circumstances associated with to-be-evaluated system limits and failures (ISO 26262).

A related criterion for the evaluation of automated vehicle HMIs is foreseeable *misuse* of the vehicle automation (Naujoks & Totzke, 2014). For example, an L2-system could promote overreliance on the automated vehicle's capabilities and engagement in distracting activities during the drive not in line with the manufacturer's intentions (Banks, Eriksson, O'Donoghue, & Stanton, 2018). In the same vein, an L3-system could appear for a user to be suitable for sleeping behind the wheel or driving under the influence of alcohol, drugs or medication (Abraham et al., 2017; Wiedemann et al., 2018). Compliance with the checklist items, which demand an efficient and intuitive communication of the system mode to drivers, might lower the possibility of misuse. Nevertheless, empirical assessments might be necessary for a comprehensive assessment of this topic.

One limitation of the proposed guidelines is that they are not intended for an evaluation of the user interaction architecture beyond the requirement that non-intended activation and deactivation should be prevented. Different automated driving systems will probably vary in interaction design and to date (Mirning et al., 2017), is not quite clear which design is the most suitable. There is a need for significantly more research in this area before guidelines on the interaction architectures can be provided.

Another limitation of the guidelines is that they primarily focus on ADS that still require occasional user intervention and therefore require a cockpit that is somewhat similar to manually driven vehicles. However, with the declining responsibility of the driver, future ADS may go along with new and radically different concepts of the vehicle interior that will likely go beyond the scope of the proposed guidelines.

5.3. Outlook and next steps

It should be noted that usability problems arising during initial and continued use of an automated driving system (Brockmann, Allgaier, Timofeev & Becker, 2016) cannot be fully predicted by relying on expert assessments – such as the one proposed in this paper – alone. The aim of the proposed method is thus not to replace usability testing with participant samples, but to complement empirical approaches with an economical evaluation tool, as both approaches might unveil different kinds of usability issues that are not easily detected by employing only one methodology (Jeffries et al., 1991).

However, a standardized and commonly accepted test protocol that can be followed when conducting evaluation studies with real users to verify the HMI of ADS is not available yet. Important details of such a test protocol, such as the selection of participants (Körber, Weißgerber, Kalb, Blaschke & Farid, 2015), instructions about the ADS (Körber, Baseler & Bengler, 2018), use cases (Naujoks, Wiedemann, Schömig, Jarosch, & Gold, 2018), traffic scenarios (Gold, Körber, Lechner & Bengler, 2016) and dependent measures (Naujoks et al., 2018) have to be specified, but their influence on test results is not yet fully understood. Significantly more research has to be conducted in this area before a thoroughly designed and robust test protocol can be established.

6. Conclusions

In this paper, we presented a compilation of most relevant standards, guidelines and best practices that can be used for the design and verification of automated vehicle HMIs. They are shared through this publication as a means of (a) stimulating discussion within the larger scientific and technical community towards future standards that may be appropriate for guiding the development of automated vehicle HMIs and (b) facilitating more rapid convergence within the technical community towards an agreed-upon set of robust guidelines and verification methods that can serve the industry in the important evolution of automation within vehicles. With the introduction of ADS rapidly approaching, both guidelines and verification methods are urgently needed to support a safe introduction of ADS.

However, it should be viewed as a starting point, and one that may still contain gaps where research in the field is still emerging. It expresses one set of guidelines and is intended as a first step towards guideline development and vehicle-verification methods. As the proposed guidelines focus on the design of the HMI, it must also be highlighted that related human factors issues such as the controllability of system limits and the foreseeable misuse of ADS are not fully covered and probably need to be verified separately, most likely in the context of comprehensive user studies.

Appendix A. Extended checklist

In the following, an extended checklist is presented that can be used in expert-based on-road tests (Table 3). The decision of whether the HMI is complying with the proposed guidelines is supported by short summaries of best practices and standards associated with the respective guideline.

Table 3

Extended checklist items.

1	Unintentional activation and deactivation should be prevented + System design ensures driver readiness before transfer of control (e.g., pushing of two buttons simultaneously, need to have both hands on the steering wheel, need to have eyes on the road, etc.) – Surprising or inexplicable driver-initiated activation/deactivation during regular use	Assessment ● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
2	The system mode should be displayed continuously + Minimum set of mode indicators present (1) functioning properly (2) currently engaged in an automated driving mode (3) currently unavailable for automated driving (4) experiencing a malfunction (5) requesting a control transition from the automated driving system to the operator – Indicators missing – Indicators not distinguishable from each other – Indicators only displayed for short periods of time – Mode indication discontinued (e.g., through pop-ups)	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
3	System state changes should be effectively communicated + Recognizable change of pictorial indicator + Auditory/haptic feedback + Communication of responsibility (e.g., by disclaimer) + Pop-up messages + Error messages are provided in case (e.g., failed activation) + Delayed reaction to control input displayed in HMI	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
4	Visual interfaces used to communicate system states should be mounted to a suitable position and distance. High-priority information should be presented close to the driver's expected line of sight + Important information displayed in 30° cone about normal line of sight + Safety-critical information displayed in 20° cone about normal line of sight + Peripheral displays support noticing of mode changes (e.g., movement or size of displays) + Status information mirrored on NDRT device	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
5	HMI elements should be grouped together according to their function to support the perception of mode indicators – Unnecessary glances to retrieve information from display (e.g., to interpret a symbol and perceive accompanying text) + Indicators pertaining to the automation are grouped together + High priority messages are easily distinguished from low-priority messages	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
6	Time-critical interactions with the system should not afford continuous attention – Important information is displayed too shortly (e.g., only for a few seconds) – While the driver is responsible for the DDT, sustained attention (longer than 1.5 s) is needed to accomplish an interaction	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
7	The visual interface should have a sufficient contrast in luminance and/or colour between foreground and background + Sufficient colour and/or luminance contrast to identify different automation modes	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
8	Texts (e.g., font types and size of characters) and symbols should be easily readable from the permitted seating position + Displayed text and symbols are big enough to be easily readable + Display resolution is good enough to be easily readable + Character width and stroke width appear to be appropriate + Text-fonts are easily readable	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
9	Commonly accepted or standardized symbols should be used to communicate the automation mode. Use of non-standard symbols should be supplemented by additional text explanations + Commonly accepted or standardized symbols are used + Non-standard symbols are supplemented with a text label + The symbols are representative for the responsibility of the driver (e.g., displaying hands on a steering wheel in case of a hands-on request)	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
10	The semantic of a message should be in accordance with its urgency + Use of notification-style to present non-critical information + Use of command-style to present critical information + Wording in accordance with criticality of the situation (e.g., “caution”, “danger”, “warning”)	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
11	Messages should be conveyed using the language of the users (e.g., national language, avoidance of technical language, use of common syntax)	

	+ Use of national language + Use of simple language + Avoidance of abbreviations + Displaying functionality rather than SAE/NHTSA/BAST-level	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
12	Text messages should be as short as possible + Messages are as short as possible + Not more than four chunks of information are displayed	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
13	Not more than five colours should be consistently used to code system states (excluding white and black) + Colours are used consistently throughout an automated driving mode + Not more than five colours are used	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
14	The colours used to communicate system states should be in accordance with common conventions and stereotypes + Colours are in accordance with common stereotypes of the user population + Red = imminent danger, yellow/amber = caution, green = hazard-free operating state	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
15	Design for colour-blindness by redundant coding and avoidance of red/green and blue/yellow combinations + green/red and yellow/blue combinations are avoided + system states are redundantly coded in a suitable way	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
16	Auditory output should raise the attention of the driver without startling her/him or causing pain Generic auditory output + suitable length (100–500 ms) + suitable loudness (50–90 dB, should be 15 dB above background noise) + frequencies between 500 and 4000 Hz Vibrotactile output + suitable length (50–200 ms) + comfortable stimuli 15–20 dB above threshold + frequencies between 150 and 300 Hz	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary ● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
17	Auditory and vibrotactile output should be adapted to the urgency of the message Generic auditory output + Auditory output of varying urgency is distinguishably different by pulse rate, frequency or loudness + Low-priority information is either unobtrusive or without auditory output Vibrotactile output + Vibrotactile output of varying urgency is distinguishably different by pulse rates, intensities, vibrating area, etc. + Different information is coded through a variation of location and timing, not frequency and amplitude	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary ● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
18	High-priority messages should be multimodal + high-priority information is presented in more than one modality + auditory or vibrotactile stimuli are also visually presented	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
19	Warning messages should orient the user towards the source of danger + warning messages lead to an orienting response to the source of danger, causing the driver to look in the direction of the hazard + warning messages to not focus the driver's attention to a display	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary
20	In case of sensor failures, their consequences and required operator steps should be displayed + unavailability of sub-systems because of sensor degradation is displayed + consequences of sensor degradation are displayed + required operator behaviours is displayed	● Major concerns ● Minor concerns ● No concerns ● Measurement necessary

We propose the outcome of the assessment to be given in four rating categories by each of the expert raters:

- “major concerns”: definite non-compliance with guideline
- “minor concerns”: partial fulfillment of guideline, but some aspects of the HMI are non-compliant
- “no concerns”: compliance of all HMI aspects with guideline

Table 4
Problem reporting sheet.

Nbr.	Description	Assessment
	Heuristic that is not met: _____	● Major concerns ● Minor concerns ● No concerns
	Use-Case: Description of AD Mode or transition:	● Measurement necessary
	Description of identified usability problem:	
	Possible solution:	

- “measurement necessary”: no definite conclusion can be given on the basis of the checklist and empirical testing is needed; this may be the case when very innovative designs are used that are not covered by current standards and best practices.

Whenever one of the guidelines is not met, the mismatch should be reported using the problem reporting sheet in Table 4, including a documentation of the use case (i.e., AD mode or transition), a description of the identified issue and (if possible) a design solution.

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